

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: D1+TD1

Date of Exam: 21.04.2022

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000226

Q1. Suppose the sender wants to send the word "SCOPE". In ASCII the five characters are coded as 01010011 01000011 01001111 01010000 01000101. Now use 2-dimensional parity and find out the extra parity bits. Discuss any two error detection scenarios for this new parity matrix. 10 M

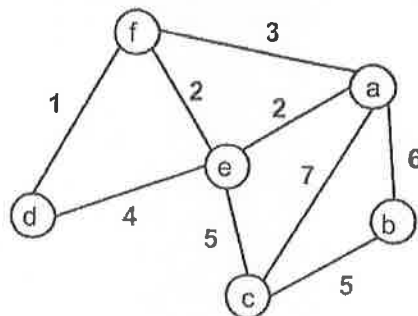
Q2.

- a) What is the main drawback of bytes stuffing? How bits stuffing is resolving this problem? Explain with an example. 05 M
- b) A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. What is the minimum frame size required to transmit in the network to detect the collision? 05 M

Q3. What is hidden terminal problem? How CSMA/CA resolve this problem and why it is best suited protocol in wireless medium? Explain it in detail. 10 M

Q4. Suppose you required 30 hosts in each network after subnet. Subnet the Class C IP Address 206.16.2.0 to fulfil your requirements. How many subnets you will create? What is the subnet mask of this network? What is the network and broadcast addresses of each subnet? 10M

Q5. Consider the following network graph with six nodes (Routers: a, b, c, d, e, f) and connected edges with weights. From node **a** and **e** calculate the shortest path spanning tree graph and table for rest of the nodes using link state routing algorithm. 10M



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q2	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q3	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q4	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10
Q5	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: D2+TD2

Date of Exam: 21.04.2022

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000227

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. A CRC is constructed to generate 3-bit CRC check bits for a 10-bit message. The generator polynomial is $X^3 + X + 1$.

- (i) Encode the data bit sequence 1001010110 (leftmost bit is the least significant) using the generator polynomial and give the codeword.
- (ii) Now assume that 5th bit (counting from the Least Significant Bit) in the codeword is changed from 1 to 0 during transmission. Now use the CRC detection algorithm and detect the error.

10 M

Q2.

- a) Is PPP protocol better than HDLC protocol for WAN communication? If yes why? List all the extra features that not available on HDLC.
- b) Suppose in a network communication, the bandwidth of the line is 2.5 Mbps and Round-Trip Time (RTT) is 25 msec. If you transmit a packet size of 2 KB, then what is the efficiency under stop and wait protocols.

05 M

05 M

Q3. What is the main reason of using back-off algorithm in CSMA/CD protocol? Explain how random amount time is calculated after collision detection for next retransmission transmission.

10 M

Q4. XYZ Company would like to subnet its network so that there are 4 separate subnets. They will need 60 computers in each subnet. Complete each of the following:

Subnet mask: 255.255.255.____

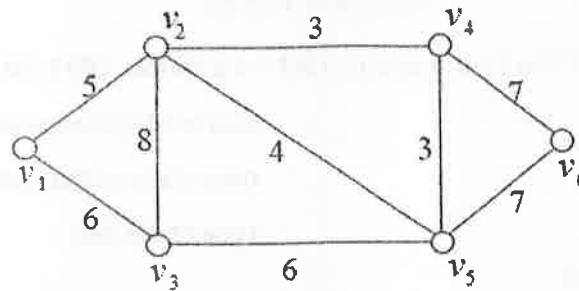
10 M

Subnets	Network Address	Host Addresses	Broadcast Address
1 st Subnet	205.100.20.____	205.100.20.____ - 205.100.20.____	205.100.20.____
2 nd Subnet	205.100.20.____	205.100.20.____ - 205.100.20.____	205.100.20.____
3 rd Subnet	205.100.20.____	205.100.20.____ - 205.100.20.____	205.100.20.____
4 th Subnet	205.100.20.____	205.100.20.____ - 205.100.20.____	205.100.20.____

Q5.

Consider the following network graph with six nodes (Routers: $v_1, v_2, v_3, v_4, v_5, v_6$) and connected edges with weights. For node V_2 and V_3 calculate the final routing table using distance vector algorithm.

10 M



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q2	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q3	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q4	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10
Q5	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: B1+TB1

Date of Exam: 19.04.2022

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000228

Q1. Consider the use of Hamming code to send 8-bit messages. Specifically, consider the message 10101011. First, calculate the Hamming codeword for this message using even parity. Second, invert one message bit of 4th position in that Hamming codeword (to represent a 1-bit error) and show how the recipient can use the check bits to correct the inverted bit. 10 M

Q2.

- a) Is slotted ALOHA protocol better than pure ALOHA protocol for WLAN communication? If yes why? List all the extra features that not available on pure ALOHA. 05 M
- b) A channel has a bit rate of 4 kbps and one-way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 50%, what is the minimum frame size. 05 M

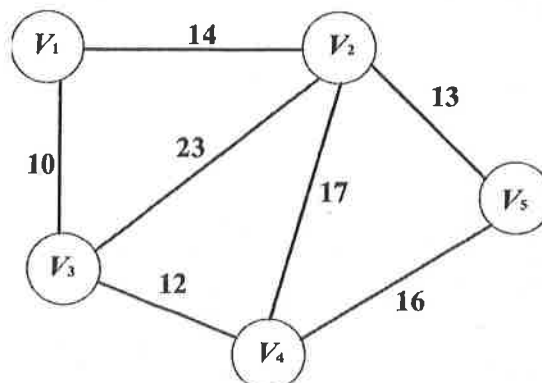
Q3. Suppose in a medium two signals are collided. How sender know about the collision signal? What is the minimum transmission time for the sender to detect the collision signal? Explain it in the context of CSMA/CD. 10 M

Q4. a) What is the network ID, broadcast address, first host IP, last host IP and number of valid IP addresses on the subnetwork that the node 199.123.12.0/27 belongs to? 5M

b) A router receives a packet with the destination address 172.109.97.32. Show how the router finds the network address of the packet. 5M

Q5.

Consider the following network graph with five nodes (Routers: v_1, v_2, v_3, v_4, v_5) and connected edges with weights. For node V_2 and V_3 calculate the final routing table using link state algorithm. 10 M



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q2	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q3	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q4	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10
Q5	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Slot: D1

Date of Exam: 21/04/2022

Duration: 90 min

Total Marks: 50

Class ID: AP2021225000229

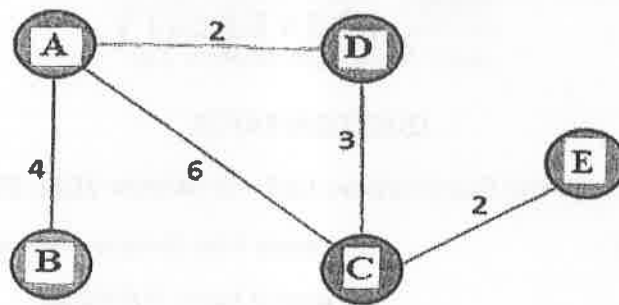
Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1.** Assume a network consisting of 4 stations A, B, C, and D, which are connected in a ring topology. Suppose if anyone station fails then remaining all the stations won't receive a connection. Give proper suggestions to overcome these link failures with a proper illustration. **(5M)**
- Q2.** Assume that you received the following data from the physical layer "11010111010111111001000111011010" to the data link layer. All the received data will undergo design issues in the data link layer process. Use the data from the physical layer and illustrate all the design issue processes in the data link layer in order to provide service to the network layer. (Note:- ESC -> 11010). **(5M)**
- Q3.** Andhra wants to send some data to Telangana using the stop and wait ARQ technique. Assume Andhra normally uses reliable transmission for data transfer and there is no frame loss. Andhra and Telangana belong to the same data format. Frames are of size 2000 bytes and the transmission rate at the sender (Andhra) is 45Kbps (1kbps = 1000 bits/sec) size of the acknowledgment is 80 bytes and the transmission rate at the receiver is 10kbps. The one-way propagation delay is 120 msec. what is the sender throughput in bytes/sec? **(10M)**
- Q4.** Consider message 11010111010110 is transmitted from host A to host B, before sending to the next level verify whether the message contains any error or not. Use encode the bits $p(x) = x^4 + x + 1$ for both sender and receiver. **(10M)**
- Q5.** Our Institution has planned to establish a digital library in AB1 and CB. It is determined that addresses in the form of 172.102.x.x.
1. How many bits are allocated for hosts?
- The IP address of the subnet in the AB1 is 172.102.43.128/26 and the IP address of the subnet in the CB is 172.102.43.192/26.
- Answer the following
2. Determine whether the address 172.102.43.213 belongs to which subnet.
 3. Find the subnet mask of the subnet in the first floor.
 4. Determine the broadcast address of both subnets.

5. Find out whether 172.102.43.100 belongs to any of the subnets. (10M)

Q6. Construct the shortest path using the distance vector routing algorithm from A to E.



QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	5	1	1	1	5
Q2	3	2	3	1	1	5
Q3	3	3	3	2	2	10
Q4	3	8	4	2	2	10
Q5	4	6	4	4	3	10
Q6	4	7	4	4	3	10



QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: D1+TD1

Date of Exam: 21.04.2022

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000231

Type your questions here

1. A sender needs to send the following: 1001100111100011. Consider 8-bit checksum is used.
 - (a) Find the checksum and the data transmitted at the sender site.
 - (b) Suppose data is received at the receiver site with no error. How this will be checked at the receiver site? [5+5]
2. Host A wants to send 9 frames to host B. The hosts agreed to go with selective repeat ARQ. How many numbers of frames are transmitted by host A if every 6th frame that is transmitted by host A is either corrupted or lost. What would be the number of transmissions in Go-Back-3 ARQ. Show all the packets on the timing diagram. [10]
3. Illustrate the concept of vulnerable time in ALOHA and slotted ALOHA protocols using a suitable diagram. A pure ALOHA network transmits 100-bit frames on a shared channel of 100 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second? [10]
4. Suppose the given IPv4 address is classless, and it is 72.16.17.0/24. What is the subnet mask of this network? You have to divide the network into 3 subnets having 128, 64, and 64

IPv4 addresses. Find the following for each of the subnets: (i) subnet address (ii) first host address (iii) last host address (iv) broadcast address (v) range of addresses. [10]

5. A datagram of 5000 byte (60 B header + 4940 B IP payload) reached at a router and must be forwarded to a link having MTU of 1000B. How many fragments will be generated? Also write MF (More Fragment), offset, and total length value for all the fragments. [10]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5			10
Q2	3	CO3	1, 2, 3, 4, 5			10
Q3	3	CO3	1, 2, 3, 4, 5			10
Q4	4	CO4	1, 2, 3, 4, 5			10
Q5	4	CO4	1, 2, 3, 4, 5			10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: D2+TD2

Date of Exam: 21.04.2022

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000232

Type your questions here

1. (a) Illustrate the procedure for pure ALOHA protocol using a suitable diagram. [5]
(b) The stations on a wireless ALOHA network are a maximum of 300 km apart. If we assume that signals propagate at 3×10^8 m/s. Find the value of T_B (Back-off time) for values of $K=2, 3, 4$. [5]
2. (a) Illustrate the process of CRC encoder and decoder using a suitable diagram. [5]
(b) Suppose the CRC divisor is 1101 and the data is 1001010. Use CRC method to find out what is the actual bit string transmitted from the sender end. [5]
3. Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 4) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If every 6th packet that A transmits gets lost (but no ACK from B ever get lost), then what is the number of packets that A will transmit for sending the message to B. Note that all the packets must be shown on the timing diagram. [10]
4. Suppose the given IPv4 address is classless, and it is 60.16.116.0/24. What is the subnet mask of this network? You have to divide the network into 4 subnets having 128, 64, 32, and 32 IPv4 addresses. Find the following for each of the subnets: (i) subnet address (ii) first host address (iii) last host address (iv) broadcast address (v) range of addresses. [10]

5. A datagram of 3000 byte (30 B header + 2970 B IP payload) reached at a router and must be forwarded to a link having MTU of 750B. How many fragments will be generated? Also write MF (More Fragment), offset, and total length value for all the fragments. [10]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5			10
Q2	3	CO3	1, 2, 3, 4, 5			10
Q3	3	CO3	1, 2, 3, 4, 5			10
Q4	4	CO4	1, 2, 3, 4, 5			10
Q5	4	CO4	1, 2, 3, 4, 5			10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot: B2+TB2

Date of Exam: 19.04.22

Duration: 90 min

Total Marks: 50

Class ID - AP2021225000233

Type your questions here

1. (a) Illustrate the transition phases of Point-to-Point protocol (PPP) using suitable diagram. [5]
(b) Assume the sender sends the dataword 10011. Suppose a simple parity check method (even parity) is used to generate the codeword that will be sent from the sender site. What codeword is sent to the receiver? Also examine the following cases at the receiver site: (i) No error occurs (ii) One single bit error changes a_1 (iii) One single-bit error changes r_0 (iv) An error change r_0 and a second error changes a_3 (v) Three bits— a_3 , a_2 , and a_1 —are changed by errors. [5]
2. Illustrate the collision mechanism in CSMA using a suitable diagram and differentiate between non-persistent, 1-persistent, and P-persistent CSMA protocols. A slotted ALOHA network transmits 100-bit frames on a shared channel of 100 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second? [5+5]
3. Station A needs to send a message consisting of 8 packets to station B using a sliding window (window size 2) and Go-Back-N error control strategy. All packets are ready and immediately available for transmission. If every 4th packet that A transmits gets lost (but no ACK from B ever get lost), then what is the number of packets that A will transmit for sending the message to B. Note that all the packets must be shown on the timing diagram. [10]
4. A datagram of 4000 byte (40 B header + 3960 B IP payload) reached at a router and must be forwarded to a link having MTU of 1000B. How many

fragments will be generated? Also write MF (More Fragment), offset, and total length value for all the fragments. [10]

5. Suppose the given IPv4 address is classless, and it is 14.6.10.0/24. What is the subnet mask of this network? You have to divide the network into 5 subnets having 128, 64, 32, 16, and 16 IPv4 addresses. Find the following for each of the subnets: (i) subnet address (ii) first host address (iii) last host address (iv) broadcast address (v) range of addresses. [10]

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5			10
Q2	3	CO3	1, 2, 3, 4, 5			10
Q3	3	CO3	1, 2, 3, 4, 5			10
Q4	4	CO4	1, 2, 3, 4, 5			10
Q5	4	CO4	1, 2, 3, 4, 5			10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Class id: AP2021225000234

Slot: D1+TD1

Date of Exam: 21.4.2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Answer all the questions

Q1. Give the relationship between dataword and codeword ? Trace the path of following data words from source to destination and perform error correction.

- a) Dataword = 0101 received at destination as 0101010.
- b) Dataword = 1110 received at destination as 1110110.
- c) Dataword = 1010 received at destination as 1010001 (10M)

Q2. A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted? Also show how error is detected at the receiver side (10M)

Q3 List the Generating rules for Walsh Tables ? Find the chips for a network with 4 stations ? show how receiving station can get data sent by the a specific station with an example? (10M)

Q4 We have a big single network having IP Address 200.1.0.0/16. Perform subnetting with the following requirements

- a) 32 customers with 64 addresses
- b) 64 customers with 128 addresses
- c) 128 customers with 256 addresses.

Design subnets and list how many address are still left with neat diagram. (10M)

Q5.

- a) List the responsibilities of network layer? List the factors that impact network layer performance?
- b) What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s (10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
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Q1	3	3	1,2,3,4,5,6,7	-		10
Q2	3	3	1,2,3,4,5,6,7	-		10
Q3	3	3	1,2,3,4,5,6,7			10
Q4	4	4	1,2,3,4,5,6,7, 8	2,3		10
Q5	4	4	1,2,3,4,5,6,7, 8	2,3		10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE 3003

Course Title: Computer Networks

Slot: D1 + TD1 (AP2021225000239)

Date of Exam:

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces

- i) 1000 frames per second?
- ii) 500 frames per second?
- iii) 250 frames per second?

[10 Marks]

Q2. Consider that, in a stop-and-wait ARQ system, the bandwidth of the line is 2 Mbps, and 1 bit takes 30 ms to make a round trip.

- i) What is the bandwidth-delay product?
- ii) Assume that the system data frames are 1000 bit in length. what will be the utilization percentage of the link if we have a protocol that can send up to 10 frames before stopping and worrying about the acknowledgements?

[10 Marks]

Q3. The sender wants to send the dataword 1001. It creates a codeword using Hamming code (even parity) and sends to the receiver. The receiver receives the codeword 1101100. Determine whether the received codeword is corrupted or not. If there is any error, also determine the bit position where the error has occurred.

[10 Marks]

Q4. An ISP is granted a block of addresses starting with 191.116.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- i) The first group has 128 customers; each customer needs 64 addresses.
- ii) The second group has 64 customers; each customer needs 256 addresses.
- iii) The third group has 128 customers; each needs 128 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

[10 Marks]

Q5. Explain the three steps involved in sending a message from a source machine to a destination machine using virtual-circuit approach for packet switching. Also, include the

creation of entries in the forwarding tables for each intermediate routers during the setup phase. [10 Marks]

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	1,2,3,4,5,6,7	-		10
Q2	3	3	1,2,3,4,5,6,7	-		10
Q3	3	3	1,2,3,4,5,6,7			10
Q4	4	4	1,2,3,4,5,6,7,8	2,3		10
Q5	4	4	1,2,3,4,5,6,7,8	2,3		10

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE 3003

Course Title: Computer Networks

Slot: D2 + TD2 (AP2021225000240)

Date of Exam:

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces

- i) 1000 frames per second.
- ii) 500 frames per second.
- iii) 250 frames per second.

[10 Marks]

Q2. Assume that the sender and the receiver uses cyclic redundancy check (CRC) for error detection. The sender sends the dataword 1101. The size of the codeword created from this dataword is of 7 bits. The divisor used for the CRC division by the encoder and decoder is 1011.

- i) Obtain the codeword created by the encoder using CRC division.
- ii) Assume the receiver receives the following codeword: 1101010. Compute the syndrome values using CRC division at the receiver end. Determine whether there is any error in the received codeword or not.

[10 Marks]

Q3. Consider that, in a stop-and-wait ARQ system, the bandwidth of the line is 2 Mbps, and 1 bit takes 30 ms to make a round trip.

- i) What is the bandwidth-delay product?
- ii) Assume that the system data frames are 1000 bit in length. what will be the utilization percentage of the link if we have a protocol that can send up to 5 frames before stopping and worrying about the acknowledgements?

[10 Marks]

Q4. A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/26.

What is the first address in the block?

What is the last address in the block?

What is total number of addresses in the block?

[5 Marks]

i) Find the class of each address:

[5 Marks]

10100011 10101010 01010101 11001100
11001100 10101010 01010101 11001100

224.128.26.38
192.168.10.43
80 A6 09 FF

Q5. An ISP is granted a block of addresses starting with 191.116.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- i) The first group has 128 customers; each customer needs 64 addresses.
- ii) The second group has 64 customers; each customer needs 256 addresses.
- iii) The third group has 128 customers; each needs 128 addresses.

Design the subblocks and find out how many addresses are still available after these allocations.

[10 Marks]

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	1,2,3,4,5,6,7	-		10
Q2	3	3	1,2,3,4,5,6,7	-		10
Q3	3	3	1,2,3,4,5,6,7			10
Q4	4	4	1,2,3,4,5,6,7,8	2,3		10
Q5	4	4	1,2,3,4,5,6,7,8	2,3		10



QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Slot: G1+TG1

Class ID: AP2021225000241

Date of Exam: 25/04/2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Answer all the questions

Q1. Datalink control protocols can either connectionless or connection oriented. If so, then what type of protocol is PPP? Elaborate the services provided by PPP with appropriate diagrams. (10 Marks)

Q2. Given the data-word 101001111 and the divisor 10111. Use CRC to find out the below mentioned questions. (10 Marks)

- a. What is the actual bit string transmitted?
- b. Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?

Q3. Assume that a stream of 4 bits (1010) is being transmitted from sender to receiver. Encode this binary bit into even pair Hamming code. (10 Marks)

Find out

- a. The total number of redundant bits?
- b. The actual information received by the receiver.
- c. Assume, there is an error at the 5th bit of the transmitted data. Mention how will the receiver correct this error.

Q4. What is the difference between Pure Aloha and slotted Aloha? Prove that the maximum throughput of Pure Aloha is $\frac{1}{2}$. Assume that a network uses pure Aloha protocol to transmit 500-bit frames through a shared channel of 500 Kb/s. Find out the throughput if the system produces (10 Marks)

- a. 1000 frames/second
- b. 750 frames/second

Q5. Consider the IP addresses enlisted below (10 Marks)

- a. 192.168.10.0/28
- b. 172.168.1.0/17

Find out:

- a. Mention the subnet masks for their respective IPs
- b. Total number of subnets for the respective IP addresses
- c. Total number IP address
- d. Total number of Hosts
- e. The broadcast id for each of the IP addresses individually

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2,4			10
Q2	3	3	2,4			10
Q3	3	3	2,4			10
Q4	3	3	2,4			10
Q5	4	4	2,4			10



QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Network

Slot: E2+TE2

Date of Exam: 22-04-2022

Duration: 90 min

Total Marks: 50

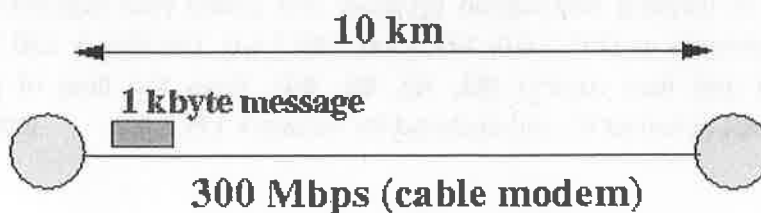
Class Id-AP2021225000242

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Given the data word 10100111 and the divisor 10111. Show the generation of the codeword at the sender site using Cyclic Redundancy Check. Show the checking of the codeword at the receiver site. **(10M)**

Q2. Consider the following transmission link during stop and wait protocol: **(10M)**

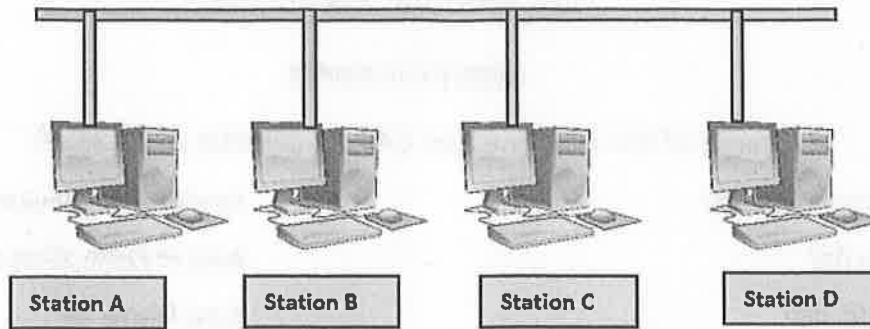


$$\text{Signal speed} = 2 \times 10^8 \text{ m/sec}$$

Calculate the following things

- a) Propagation time (2)
- b) Calculate the useful time and Useless time (4)
- c) Link Utilization (4)

- Q3.** Consider a Scenario of CSMA/CD for the following architecture. Station A starts transmitting the data packets, before it ends, station D Starts transmitting its data packets. At some point during there is a collision. Derive the condition for detecting the collision in CSMA/CD. **(10M)**



- Q4.** An organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.
- Find the subnet mask. (2)
 - Find the number of addresses in each subnet. (2)
 - Find the first and last addresses in subnet 1. (4)
 - Find the first and last addresses in subnet 32. (2)
- (10M)**
- Q5.** Explain the working of Routing information protocol and Justify your explanation by considering seven networks as (195.2.4.0, 195.2.5.0, 195.2.6.0, 130.10.0.0, 130.11.0.0, 205.5.5.0, 205.5.6.0) and four routers (R1, R2, R3, R4). Show the flow of packet numbered 1 that arrives at router R1 and destined for network 195.2.6.0. **(10M)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	2,4	-	-	10
Q2	3	3	2,5	-	-	10
Q3	3	3	2,4	-	-	10
Q4	4	4	2,4	-	-	10
Q5	4	4	2,4	-	-	10

QUESTION PAPER

Name of the Examination: CAT (Long summer 2021-22)

Course Code: CSE3003

Course Title: Computer Networks

Slot: E+TE+TEE

Date of Exam: 06-07-2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1. a.** Find out an OSI layer that carries out the encryption at the transmitter and decryption at the receiver and also explain the other important functions of that layer. **4M**
- b.** Explain in detail about a network topology in which the total number of ports required by each device is $N-1$ where N is the number of devices in the network. Moreover, list the merits and demerits of that topology. **6M**
- Q2. a.** Identify the Cyclic Redundancy Check (CRC) using the following polynomial and explain error detection in detail. Here, the message polynomial is $m(x) = x^6 + x^4 + 1$ and the generator polynomial is $g(x) = 1001$. **8M**
- b.** Examine Vertical Redundancy Check (VRC) in detail for the message 000001 **2M**
- Q3.** Analyse in detail about a MAC protocol which requires that each station first check the state of the medium before sending. Moreover, briefly explain the modified version of that protocol which has been specially designed for wireless networks. **10M**
- Q4. a.** Analyse in detail about the various types of a media where the transmission of electromagnetic waves is done without using any of the physical transmission media. **6M**
- b.** Explain in detail about the connection-oriented switching technique in terms of their message forwarding strategy. **4M**
- Q5. a.** Survey the frame structure of a traditional technology that connects devices in a wired local area network (LAN) or wide area network (WAN). Moreover, list the various classifications of that technology in detail. **4M**
- b.** Encode the binary data 1010001 using the following encoding methods **6M**
1. RZ
 2. NRZ -I
 3. Bipolar
 4. Differential Manchester

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2,3,4,5,6	-	1	10
Q2	3	3	1,2,3,4,5,6	-	1	10
Q3	3	3	1,2,3,4,5,6	-	1	10
Q4	2	2	1,2,3,4,5,6	-	1	10
Q5	2,3	2,3	1,2,3,4,5,6	-	1	10

QUESTION PAPER

Name of the Examination: CAT (Long summer 2021-22)

Course Code: CSE3003

Course Title: Computer Networks

Slot: E+TE+TEE

Date of Exam: 06-07-2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1. a.** Find out an OSI layer that carries out the encryption at the transmitter and decryption at the receiver and also explain the other important functions of that layer. **4M**
- b.** Explain in detail about a network topology in which the total number of ports required by each device is $N-1$ where N is the number of devices in the network. Moreover, list the merits and demerits of that topology. **6M**
- Q2. a.** Identify the Cyclic Redundancy Check (CRC) using the following polynomial and explain error detection in detail. Here, the message polynomial is $m(x) = x^6 + x^4 + 1$ and the generator polynomial is $g(x) = 1001$. **8M**
- b.** Examine Vertical Redundancy Check (VRC) in detail for the message 000001 **2M**
- Q3.** Analyse in detail about a MAC protocol which requires that each station first check the state of the medium before sending. Moreover, briefly explain the modified version of that protocol which has been specially designed for wireless networks. **10M**
- Q4. a.** Analyse in detail about the various types of a media where the transmission of electromagnetic waves is done without using any of the physical transmission media. **6M**
- b.** Explain in detail about the connection-oriented switching technique in terms of their message forwarding strategy. **4M**
- Q5. a.** Survey the frame structure of a traditional technology that connects devices in a wired local area network (LAN) or wide area network (WAN). Moreover, list the various classifications of that technology in detail. **4M**
- b.** Encode the binary data 1010001 using the following encoding methods **6M**
1. RZ
 2. NRZ -I
 3. Bipolar
 4. Differential Manchester

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2,3,4,5,6	-	1	10
Q2	3	3	1,2,3,4,5,6	-	1	10
Q3	3	3	1,2,3,4,5,6	-	1	10
Q4	2	2	1,2,3,4,5,6	-	1	10
Q5	2,3	2,3	1,2,3,4,5,6	-	1	10



QUESTION PAPER

Name of the Examination: CAT (Fast Track 2022-2023)

Course Code: CSE3003

Course Title: Computer Networks

Slot: B+TB+TBB

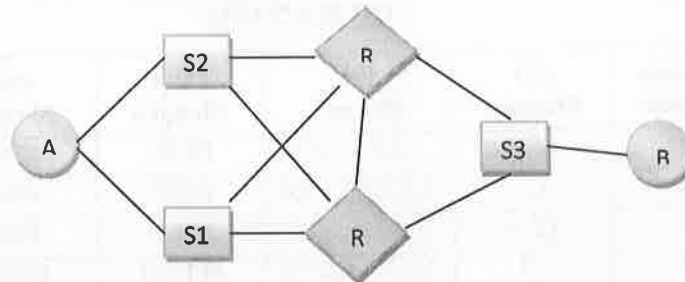
Date of Exam: 29-07-2022

Duration: 90 min

Total Marks: 50

Class Id: AP2022231000047

- Q1.** Bob transmits 3 GB of data to Alice using the TCP/IP protocol. Why is a layered architecture necessary for this data communication? Describe the exchange of information between two nodes at each tier. **(10 Marks)**
- Q2.** Consider a sample network given here:



Assume Sender A needs to send 10 packets of data to Receiver B. Explain how the following switching techniques will use the given network efficiently. **(10 Marks)**

- a) Circuit switching
- b) Packet switching

Q3. a) Draw the graph of the NRZ-L, Manchester scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level.

a. 00000000 b. 11111111

(5 Marks)

b) Sender A sends 12 frames based on the principle that the sender will not transmit the next frame until it receives acknowledgment of the last transmitted frame.

Draw the timing diagram when all frames are sent successfully.

(5 Marks)

Q4. Consider a data word 110011.

- a) Find the codeword using CRC with generator polynomial as X^3+X+1 **(5 Marks)**
- b) If the codeword generated in the previous question is transmitted, then how can you verify the bit error at the receiver end? **(5 Marks)**

Q5. a) Show how U-Frames can be used for connection establishment and connection release. Node A asks for a connection with a set of asynchronous balanced mode; Node B gives a positive response, with an unnumbered ack. frame. After these 2 exchanges, data can be transferred between 2 nodes. After data transfer; Node A sends a DISC (disconnect) frame to release the connection; It is confirmed by node B responding with an unnumbered acknowledgment. **(5 Marks)**

b) Define the type of the following destination addresses with respect to Ethernet: **(5 Marks)**

a) 4A:30:10:21:10:1A

b) 47:20:1B:2E:08:EE

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1,2,4	1,2,3,4,5,6,7	PEO2	PSO1	10
Q2	2	1,2	1,2,3,4,5,6,7,8	PEO3	PSO3	10
Q3	2	1,2,4	1,2,3,4,5,6,7	PEO3	PSO3	10
Q4	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10

QUESTION PAPER

Name of the Examination: RE-CAT – 2 (Winter 2021-2022)

Course Code: CSE-3003

Course Title: Computer Networks

Slot:

Date of Exam:

Duration: 90 min

Total Marks: 50

Class ID -

Q1. Consider the use of Hamming code to send 8-bit messages. Specifically, consider the message 11001010. Now calculate the Hamming codeword for this message using even parity for sender side. To show error detection and correction invert one message bit of 5th position in that Hamming codeword (to represent a 1-bit error). Show how the recipient can use the check bits to correct the inverted bit. 10 M

Q2.

- a) Why pure ALOHA protocol performance is not good and how slotted ALOHA improved the performance double of pure ALOHA? 05 M
- b) A channel has a bit rate of 2 kbps and one-way propagation delay of 10 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of at least 50%, what is the minimum frame size. 05 M

Q3. How Exponential Backoff Algorithm is related to CSMA/CD protocol? Explain backoff algorithm with a suitable flow chart. 10 M

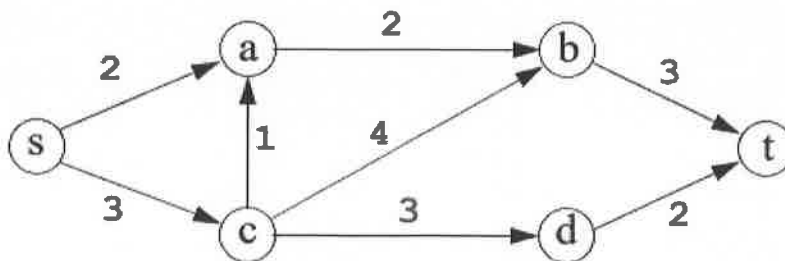
Q4. Write the subnet mask, network address, broadcast address, first usable address and valid host range for each of the subnets for the given IPv4 address.

192.168.100.0, with 2 bits of subnetting

10M

Q5.

Consider the following network graph with five nodes (Routers: S, a, b, c, d, t) and connected edges with weights. For node **b** and **c** calculate the final routing table using link state algorithm. 10 M



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q2	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q3	3	CO3	1, 2, 3, 4, 5	PEO3	PSO3	10
Q4	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10
Q5	4	CO4	1, 2, 3, 4, 5	PEO4	PSO4	10



QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Slot: A+TA1

Date of Exam: 23-05-2022

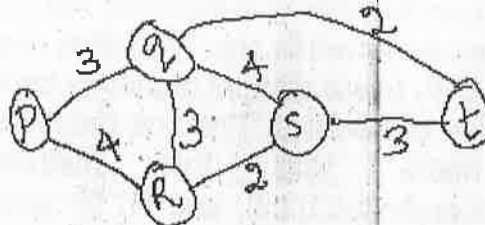
Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1.** Apply Link state routing and find the shortest path from all the routers to all other routers and draw the unicast routing path to all other nodes for the network graph given below. **(10 Marks)**



- Q2.** The following is a dump of a UDP header in hexadecimal format. EC84 0011 001A 001D

- a) What is the source port number? **(2 Marks)**
- b) What is the destination port number? **(2 Marks)**
- c) What is the total length of the user datagram? **(2 Marks)**
- d) What is the length of the data? **(2 Marks)**
- e) What is the client process? **(2 Marks)**

- Q3.** Consider a TCP communication between the Sender X and Receiver Y. Sender X have 30 segments, cwnd=15 and rwnd=15. The congestion control in the network sets the slow start threshold level as 10 and this will be reduced by half when congestion occurs. Explain the congestion control timing diagram with window representation on the following conditions. **(10 Marks)**

- a) If 15th segment is time out
- b) If 23rd segment receives 3 Duplicate ACK
- c) If 28th segment is time out

- Q4.** Assume a sender A have 10 segments which consists of 15 data bytes each. And sender B have 15 segments which consists of 10 data bytes each. Let

both senders have started to send the data to receiver X through router R1 at the rate of 2 segments per second. Explain how the following scheduling algorithm improves the quality of service at another intermediate router R2 where it has the total outflow as 3 segments per second. (10 Marks)

- FIFO
- Priority Queue (all segments from sender B are higher priority)
- Weighted fair Queue (all odd numbered segments from sender B have weight 1, all even numbered segments from sender B have weight 2, and all segments from sender A have weight 3)

Q5. Draw the FTP communication timeline between the client and server. Assume that FTP server is ready. Client confirms the authentication with user name movie and password Hollywood. connection fails and requires password. client re-confirms the authentication with password Tollywood. data connection established and will be open shortly. now file is request as /trailer/poster1.jpg and file is requested of type image as a file structure. got response as more information needed. now file mode is responded as stream. got response with insufficient storage. now file mode is responded as compressed. now data is transferred and server sends close data connection and connection ends.

Q6. Show the HTTP communication dialogue between the host A and host B over the web page www.computernetworks.org. The initial request has been made with the method (PUT), URL (www.comnet.org/index.html), and HTTP version (1.1). There are four lines of headers. The first response is Not Found. The second request made with the method (GET), URL (www.computernetworks.org/index.html), and HTTP version (1.1). There are four lines of headers. At this state, internal error happened at server. Again, the same request has been sent. Now the host A got the response message contains the status line and four lines of headers. The created document, which is a CGI document, is included as the body. (10 Marks)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Mark s
Q1	4	3,4	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q6	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Network

Slot: B2+TB2

Date of Exam: 24/05/2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1.

- A) A receiver receives a 7-bit Hamming code of 1011011 in even parity state. Find out is there any error in received code? If there is an error then find the position of error and correct it. (6 M)

- B) For the bit stream 1010101101, describe and sketch the waveforms for each of the encoding techniques given below:

(i) Differential Manchester

(ii) NRZ-I

(4 M)

Q2.

The following is the dump of a TCP header contents in hexadecimal format. Based on this header find the answer of the following questions?

(10 M)

0322001E 00000001 00000000 500203FF 00000000

- (i) What is the source port number?
- (ii) What is the destination port number?
- (iii) What is the sequence number?
- (iv) What is the length of header?
- (v) What is the window size?

Q3.

A)

In a TCP connection, the initial sequence number at the client side is 2171. The client opens the connection, sends three segments, the second of which carries 1,000 bytes of data and closes the connection. What is the value of the sequence number in each of the following segments sent by the client?

- i. The SYN segment
- ii. The data segment
- iii. The FIN segment

(6 M)

- B) List the fields in the TCP header that are missing from UDP header. Give the reason for their absence.

(4 M)

Q4.

- A. Suppose your network traffic is bursty in nature. How you can make it regulated maximum rate traffic? (5M)
- B. Consider the size of a congestion window of a TCP connection be 36 KB when a timeout occurs. The round-trip time the connection is 200 μ sec and the maximum segment size used is 2 KB. What is the time taken by the TCP connection to get back to 36 KB congestion window? (5M)

Q5.

You are supposed to attend a meeting at Bengaluru. While making arrangements, you planned to stay in a hotel. In this regard, you search in yellow pages to get the information about all the good hotels near the meeting spot. Relate this use case to resolving the DNS. Write the process of resolving. Describe the other way of resolving the query. (10 M)

Q6. A)

FTP uses the services of TCP for exchanging control information and data transfer. Could FTP have used the services of UDP for either of these two conditions? Explain (5 M)

B)

Suppose you want to send an e-mail to your friend using a TCP connection. Now you draw a time line diagram from source to destination and mention all the phases with suitable name. (5 M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3,4	5,6	1	1	1	10
Q2	5	4	2	1	1	10
Q3	5	1	4	1	1	10
Q4	5	3	4	1	1	10
Q5	6	7	3	3	1	10
Q6	6	8	3	3	2	10

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Network

Slot: B1+TB1

Date of Exam: 24/05/2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1.

Suppose you have done subnetting for your class C network 199.108.10.0 with a subnet mask of 255.255.255.252.

Please find the following: number of sub-networks, number of hosts per network, the full range of the first three networks, and the usable address range from those first three networks. Additionally, identify the broadcast and network addresses for each network. **(10 M)**

Q2.

The following is the dump of a TCP header contents in hexadecimal format. Based on this header find the answer of the following questions?

B3910028 000000A1 000000A2 500203EE 00000000

- (i) What is the sequence number?
- (ii) What is acknowledgement number?
- (iii) What is the length of the header?
- (iv) What is the window size?
- (v) What is the type of the segment?

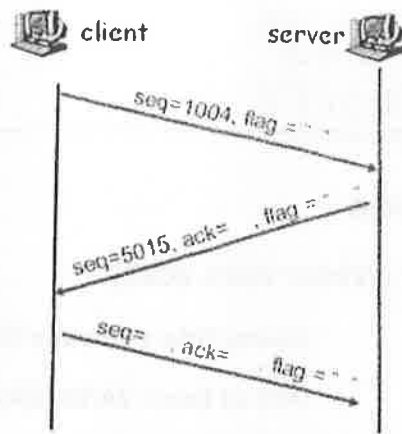
(10 M)

Q3.

An International Conference is planned by a university. Due to this pandemic situation, it has been planned to conduct virtually. Organiser want to make the network connection reliable. Determine the best suited protocol to perform this and explain the header frame format of that protocol. **(10 M)**

Q4.

- A. Suppose the TCP packet transmission between host A and host B follows the following scenario, fill in the missing sequence number and ack number. **(5M)**



TCP connection setup

- B. Distinguish between a time-out event and the three-duplicates-Acks event. Which one is a stronger sign of congestion in the network? Why? (5M)

Q5.

A) Suppose you open a start-up company "starwar" and want to set up your company network.

Your network has the following servers:

- I. DNS server: "dns1.starwar.com" with IP as "128.119.12.40"
- II. Web server: "starwar.com" with two IP as "128.119.12.55" and "128.119.12.56". The web server also has a name as "www.starwar.com".
- III. Email server: "galaxy.starwar.com" with IP as "128.119.12.60"

Your company's email address is "username@starwar.com".

- a). What resource records (RRs) do you need to provide to the upper level ".com" Registrar?
- b). What RRs do you need to put in your company's DNS server? (6 M)

B) In FTP two different ports address available, port 20 and port 21.? What is the use of these two different ports? (4 M)

Q6.

Suppose you have sent a request from your browser to the VTOP server? In this request which protocol is mainly used by the browser? Explain the details of request page of client and reply/response page of server. (10 M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3,4	5,6	1	1	1	10
Q2	5	4	2	1	1	10
Q3	5	1	4	1	1	10
Q4	5	3	4	1	1	10
Q5	6	7	3	3	1	10
Q6	6	8	3	3	2	10

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Slot: D+TD1

Date of Exam: 25-05-2022

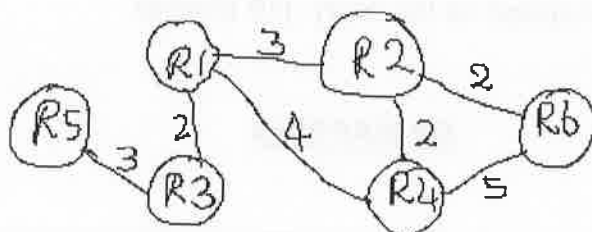
Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

- Q1.** Apply Link state routing and find the shortest path from all the routers to all other routers and draw the unicast routing path to all other nodes for the network graph given below. **(10 Marks)**



- Q2.** The following is a dump of a TCP header in hexadecimal format E2930017 00001201 00001401 401807DC 00000000

- a) What is sequence number? **(2 Marks)**
- b) What is the acknowledgment number? **(2 Marks)**
- c) What is the length of the header? **(2 Marks)**
- d) What is the type of the segment? **(2 Marks)**
- e) What is the window size? **(2 Marks)**

- Q3.** Host A communicates with Host B using the TCP connection. Assume Host B has received all bytes through 400. If host A sends 2 segments to host B back-to-back and assume the 1st and 2nd segments comprise 100 and 50 bytes respectively. The sequence number in the 1st segment is 111. Source port= 24 destination port =300, When host B receives a segment, it acknowledges it by sending an acknowledgement signal. **(10 Marks)**

- a) Show the timing diagram for connection establishment in TCP.
- b) Show the timing diagram of data transmission.
- c) Show the timing diagram for connection termination.

- Q4.** Let Host A wants to send 10 segments of data of size 1024 bytes each to Host B.
- If the host A is compromising with un-reliable data communication, what type of transport layer services can be suggested? Why? Explain your answers with a timing diagram. **(5 Marks)**
 - If the host A wants to send the data only to Host B, what is a suitable Application layer paradigm? Why? **(5 Marks)**
- Q5.** Consider, you have to send an email for leave request from your ranjan@vitap.ac.in to your boss head.sde@vitap.ac.in. Explain the email communication process with timing diagram of SMTP protocol. **(10 Marks)**
- Q6.** Show the HTTP communication dialogue between the host A and host B over the web page www.computernetworks.org. The initial request has been made with the method (GET), URL (www.facebook.org/index.html), and HTTP version (1.1). There are four lines of headers. The first response is Not Found. The second request made with the method (POST), URL (www.facebook.com/index.html), and HTTP version (1.1). There are four lines of headers. At this state, internal error happened at server. Again, the same request has been sent. Now the host A got the response message contains the status line and four lines of headers. The created document, which is a CGI document, is included as the body. **(10 Marks)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	4	3,4	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q6	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10



QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Networks

Slot: D+TD1

Date of Exam: 25-05-2022

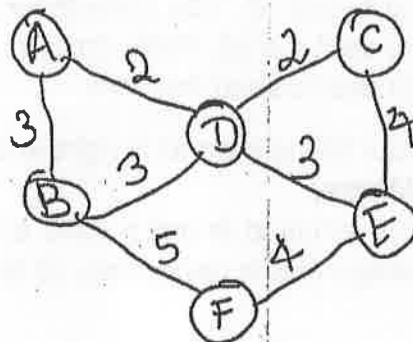
Duration: 120 min

Total Marks: 60.

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1. Apply Distance vector routing and find the shortest path from all the routers to all other routers and find the unused paths for the network graph given below. **(10 Marks)**



Q2. Consider an instance of TCP Additive Increase Multiplication Decrease (AIMD) Algorithm where the window size of the start of the slow phase is 3 MSS and the threshold of the start of the first transmission is 6 MSS. Assume that a time-out occurs during fifth transmission. Find the congestion window size at the end of 12th transmission and draw congestion control flow graph **(10 Marks)**

Q3. Host A communicates with Host B using the TCP connection. Assume Host B has received all bytes through 400. If host A sends 2 segments to host B back-to-back and assume the 1st and 2nd segments comprise 100 and 50 bytes respectively. The sequence number in the 1st segment is 111. Source port= 24 destination port =300, When host B receives a segment, it acknowledges it by sending an acknowledgement signal. **(10 Marks)**

- a) Find out the sequence number, source and destination port number in the second segment.
- b) What would be the ACK number if the first segment arrives before the second segment?

- c) Find the destination and source port number for (ii) transmission.
- d) If the second segment comes before the arrival of the 1st segment, then determine the ACK number in the ACK of the 1st arriving segment

Q4. Let Host A wants to send 10 segments of data of size 1024 bytes each to Host B.

- a) If the host ensures reliable data communication, what type of transport layer services can be suggested? Why? Explain your answers with a timing diagram. **(5 Marks)**
- b) If the host wants to send the data that needs to be distributed to all other nodes through the Host B, what is a suitable Application layer paradigm? Why? **(5 Marks)**

Q5. a) Consider, you have deployed a web application at the server with an IP address of 172.16.34.12:8080/cgpa.jsp. Explain how DNS can be used to deployed this web application over internet with .com domain. **(5 Marks)**

b) Explain how the name address resolution can be done between the IP address and the domain name. Draw the DNS message format between the client and server. **(5 Marks)**

Q6. You want to send a mail from your mail to your old friend. But you are unable to recollect your friend's mail id. You remember his name and by trial-and-error method, you have sent a mail. Write down in sequence the commands in SMTP that you need to use to send this mail.

- a) case 1: Assume your friend's email is right and the message is received at the other end. **(5 Marks)**
- b) case 2: Assume the email id is not a valid address. Then write down the suitable error message that is generated. **(5 Marks)**

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Mark s
Q1	4	3,4	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q6	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10



QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3003

Course Title: Computer Network

Slot: D2+TD2

Date of Exam: 26/05/2022

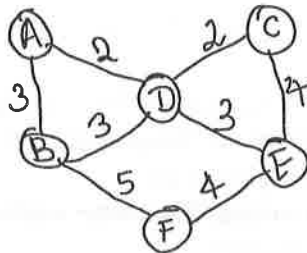
Duration: 120 min

Total Marks: 60

6 x 10 = 60 Marks

Answer all the questions

- Q1** Apply Link state vector routing and find the shortest path from all the routers to all other routers and find the unused paths for the network graph given below. **(10 Marks)**



- Q2** The following is the content of a TCP Header in hexadecimal format:
XXXXXXXX X0002101 00002000 A00107FF 00000000

Where the first 9 X's should be replaced with your registration number. Assume your registration number is **20BCE7021**, then the content of TCP header will be as follows:

20BCE702 10002101 00002000 A00107FF 00000000

In case your registration contains any letter except A, B, C, D, and E, then replace that letter with E, and then proceed as stated above.

For example, if your registration number is 20BC**N**3001, then replace the N with E. So the registration number will become 20BC**E**3001.

In this case, the contents of TCP header will be as follows:

20BCE300 10002101 00002000 A00107FF 00000000

Based on your registration number, answer the following:

- I) What is the source port number?
- II) What is the destination port number?
- III) What is sequence number?
- IV) What is the acknowledgment number?
- V) What is the length of the header in bytes?
- VI) What is the type of the segment?
- VII) What is the window size?

- Q3** The ssthresh value for a Reno TCP station is set to 24 MSS.
- Assume the station is now in the slow-start state with $cwnd = 8$ and $ssthresh = 24$ MSS. After one RTT, it receives 8 nonduplicate ACKs followed by a time-out. What will be the state of the station, value of $cwnd$ and $ssthresh$, before and after the time-out?
 - Assume the station is now in the slow-start state with $cwnd = 8$ and $ssthresh = 24$ MSS. After one RTT, it receives 8 nonduplicate ACKs followed by a 3 duplicate ACKs. What will be the state of the station, value of $cwnd$ and $ssthresh$, before and after receiving the 3 duplicate ACKs?
 - When will Reno TCP enter into congestion-avoidance state? Explain through a scenario.
- Q4** Consider a TCP connection is established between a client and a server. Assume unidirectional communication from the client to server. During the connection establishment, the client sends a SYN segment with sequence number 250. The server then sends a SYN+ACK segment containing ACK = 251 and $rwnd = 1000$. The size of send window is set to 1000. The client sends the ACK segment to complete the connection establishment. Now, the data transfer phase will start in the following sequence:
- The client sends a segment A consisting of 250 bytes. What will be the sequence number of this segment?
 - The server acknowledges the receipt of segment A by sending segment B. What will be the value of ACK and $rwnd$ in segment B? What will be the value of R_n (next byte expected to be received) after sending segment B?
 - What will be the value of S_f (the first outstanding byte waiting to be acknowledged) after the segment B is received by the client?
 - If the server process consumes 200 bytes from TCP server, what will be the value of R_n and $rwnd$?
 - The client sends a segment C consisting of 250 bytes. What will be the sequence number of this segment? What will be the value of S_n (next byte to send)?
- Q5** Explain with timeline diagrams how client-server communication happens using the HTTP protocol with the help of request and response codes. Explain how statefulness is achieved in HTTP.
- Q6** Explain how DNS resolves the queries through the iterative and recursive approaches using step-by-step flow diagram. Also, identify various components present in the DNS architecture.

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	4	3,4	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q6	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE 3003

Course Title: Computer Networks

Slot: G1 + TG1

Date of Exam: 30/05/2022

Duration: 120 min

Total Marks: 60

Instructions:

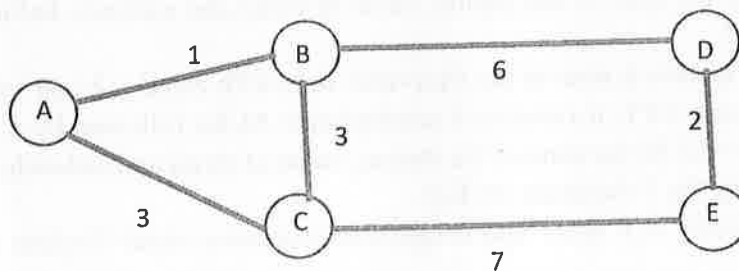
1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1.

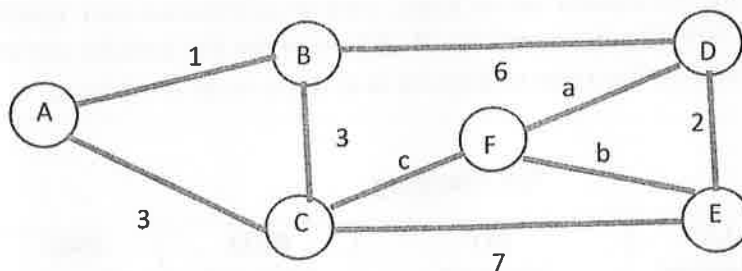
Answer all the questions

5x 12 = 60 Marks

1. Case-1: Find the shortest path from node A to every other node in the network using Distance vector routing protocol for the Figure given below. Generate the final topology. (6 Marks)



Case-2: Assume that a new node F is added to the above-mentioned topology. The links from the node F to other nodes are mentioned as a, b and c. Find out the constraints on a, b, and c so that they match the A's routing tree generated in case-1 (6 Marks)



2. Describe the several congestion control policies followed by the TCP protocol. Assume that a TCP client is expecting to receive byte 3000 from the TCP server. It also receives the segment with the sequence number 3000 that carries 250 bytes of information.

Assume that, the client further has no data to send and wants to terminate the connection, whereas the server doesn't want to terminate the connection at that movement. What will be the reaction of the TCP client? Justify your answer with a flow diagram.
(12 Marks)

3. A. The following is the part of TCP header dump in hexadecimal format
40A9 0017 00000001 00000000 500407FF 00000000 (8 Marks)

Find out the

- Source port number
- Destination port number
- Source sequence number
- Acknowledgement number
- Length of the header in bytes
- What is the type of the segment.
- Window size

B. Assume that a TCP client needs to establish connection with the TCP server from the source port 'A' to destination port '80'. Assume a random sequence number for both server and client. Consider the window size for the client to be 12000 bytes with a MSS of 600 B where as for the server the window size is 18000 with a MSS of 600 B. Set up the flags appropriately and draw the flow diagram for the connection establishment phase using TCP protocol. (4 Marks)

4. The ssthresh value for a Reno TCP station is set to 24 MSS. (12 Marks)
- Assume the station is now in the slow-start state with $cwnd = 8$ and $ssthresh = 24$ MSS. After one RTT, it receives 8 nonduplicate ACKs followed by a time-out. What will be the state of the station, value of $cwnd$ and $ssthresh$, before and after the time-out?
 - Assume the station is now in the slow-start state with $cwnd = 8$ and $ssthresh = 24$ MSS. After one RTT, it receives 8 nonduplicate ACKs followed by a 3 duplicate ACKs. What will be the state of the station, value of $cwnd$ and $ssthresh$, before and after receiving the 3 duplicate ACKs?
 - When will Reno TCP enter into congestion-avoidance state? Explain through an example.
5. Discuss with a brief diagram how a TCP/IP suite uses a DNS client and DNS server to map a name to an address. Assume that the user wants to use a file transfer client to access the corresponding file transfer server running on a remote host. The user knows only the file transfer server name, such as *afileservice.com*. In order to make a connection TCP/IP suite needs the IP address of the file transfer server to make a connection. Discuss the steps to map the host name to an IP address.
(12 Marks)

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	4	3,4	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	12
Q2	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	12
Q3	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	12
Q4	5	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	12
Q5	6	4,5,6	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	12

QUESTION PAPER

Name of the Examination: FAT (Long summer 2021-22)

Course Code: CSE3003

Course Title: Computer Networks

Slot: E+TE+TEE

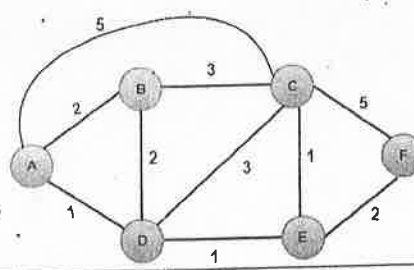
Date of Exam: 02-08-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Q1.	a.	Examine Longitudinal Redundancy Check (LRC) in detail for the message 000001 101010 001010 101100	4M
	b.	Explain in detail the Checksum method to detect the error for the following data unit 10011001 11100010 00100100 10000100	6M
Q2.	a.	On a wireless link, the probability of packet error is 25%. A stop and wait protocol is used to transfer data across the link. The channel condition is assumed to be independent of transmission to transmission. What is the average number of transmission attempts required to transfer 500 packets?	4M
	b.	Compute the fraction of the bandwidth that is wasted on overhead (headers and retransmissions) for a protocol on a heavily loaded 100 Kbps satellite channel with data frames consisting of 50 bits header and 4000 data bits. ACK frames never occur. NAK frames are 60 bits. The error rate for data frames is 0.2 and the error rate for NAK frames is negligible. Assume 100 data frames for transmission.	6M
Q3.		Consider the following network and find the best path route between the routers A (Source) and E (Destination) using link-state routing Dijkstra's algorithm. 	10M
Q4.	a.	Consider a single network having IP Address 200.1.2.0 is divided into 2 subnets. Then, find the IP address of each subnet, the total number of hosts that can be configured in each subnet, and the direct broadcast address of each subnet.	5M
	b.	Explain in detail about the protocol that assigns the IP addresses to the computers in the lease manner.	5M

Q5.	a.	Explain in detail about the transport layer protocol which is used where fast communication is required and does not care about reliability.	6M
	b.	Explain in detail the addressing mechanism which is used to deliver the data to the right process in the computer system.	4M
Q6.	a.	How many bits may be required for encoding the message 'Mississippi' using Huffman lossless compression technique?	7M
	b.	Calculate the Run Length Encoded string for the input string "wwwaaaadexxxxxx"	3M

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	1,2,3,4,5,6	-	1	10
Q2	3	3	1,2,3,4,5,6	-	1	10
Q3	4	4	1,2,3,4,5,6	-	1	10
Q4	4	4	1,2,3,4,5,6	-	1	10
Q5	5	5	1,2,3,4,5,6	-	1	10
Q6	6	6	1,2,3,4,5,6	-	1	10

CAT - 2 QUESTION PAPER

School	Computer Science and Engineering		
Course Code	CSE3003	Course Title	Computer Networks
Semester	Win 2021	Date of exam	21-04-2022
Slot	D1	Total Marks	50
Duration	90 Min	Class ID	AP2021225000236

Answer all the questions

5 x 10 = 50 Marks

Q1. Sender A sends 12 frames based on the principle that the sender will not transmit the next frame until it receives the acknowledgement of the last transmitted frame.

- Draw the timing diagram when all frames are sent successfully. (5 Marks)
- Draw the timing diagram, if every 3rd packet that is being transmitted is lost, then how many transactions are required? (5 Marks)

Q2. Consider a data word 110011.

- Find the codeword using CRC with generator polynomial as X^3+X+1 (5 Marks)
- If the codeword generated in the previous question is transmitted, then how can you verify the bit error at the receiver end? (5 Marks)

Q3. a) Consider the data to send: 3,4,2,5,6,2,8,9,4,2,5,3,2. Design a 3 variable length frame with field delimiter as its byte count. (3 Marks)

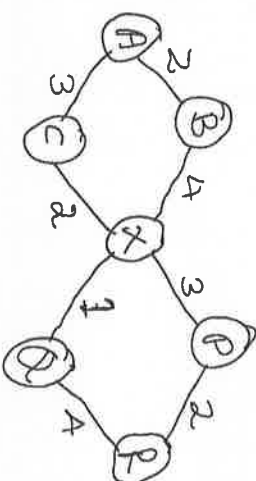
- Show how S-Frames can be used to transmit the control information when a node A wants to send 10 frames to node B. Node A sends two frames with a set of receiver ready mode; Node B gives a positive acknowledgement frame for both frames. After these 2 exchanges, data has been communicated when the receiver is in not ready mode. Hence the 3rd data is rejected. So, 3rd frame is re transmitted with receiver ready mode. Thereafter every 4th frame is selectively rejected. (5 Marks)

- An address in a block is given as 171.192.127.0. Find the number of addresses in the block, the subnet mask, the first address, and the last address. (2 Marks)

Q4. a) Suppose we have four small networks with network ID as 192.16.0.0, 192.16.1.0, 192.16.2.0, 192.16.3.0. Apply super netting and find new super net mask with CIDR suffix (5 Marks)

- Expand the given IPv6 "21: ab1: b3c4: 15:: a::" and find prefix, subnet id and interface id. (5 Marks)

Q5. Apply Distance vector routing and find the shortest path from all the routers to all other routers and find the unused links for the network graph given below. (10 Marks)



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	3,4	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10

CAT - 2 QUESTION PAPER

School	Computer Science and Engineering		
Course Code	CSE3003	Course Title	Computer Networks
Semester	Win 2021	Date of exam	21-04-2022
Slot	D2	Total Marks	50
Duration	90 Min	Class ID	AP2021225000237

Answer all the questions

5 x 10 = 50 Marks

Q1. Sender A have 10 frames to send to receiver B. Sender A have feature to send 4 frames without acknowledgment and sender needs to retransmit all the 4 frames in case of failure

- How many transactions are required, if every 5th frame is failure? Explain the answer with timing diagram. (5 Marks)
- How many transactions are required, if the acknowledgement of every 4th frame is lost? Explain the answer with timing diagram. (5 Marks)

Q2. Consider the data word 'HELLO' (Note: ASCII Code for hello is: 72, 69, 76, 76, 79)

- Find the codeword using Checksum (5 Marks)
- If the codeword generated in the previous question is transmitted in reverse order, then how can you verify the codeword at the receiver end? (5 Marks)

Q3. a) Show how PPP Protocol can be used to transmit the data when a node A wants to send 100 kb data to node B. Node A sends the first configuration request; Node B gives a negative acknowledgement. Then node A sends second configuration request which is accepted by positive configuration acknowledgement. After these 2 exchanges, data has been communicated. After data transfer; Node A sends a termination request to release the connection; It is confirmed by node B responding with a termination acknowledgement. (5 Marks)

- Explain how BGP protocol implements the route selection. (3 Marks)

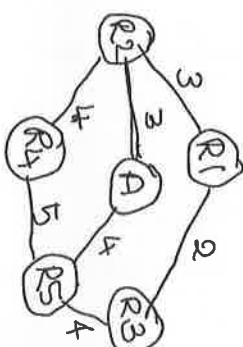
- Assume Sender A knows the receiver B's network address. Explain how node A identifies the link layer address of the receiver B in a network. (2 Marks)

Q4. You have sub-netted your class B network 172.168.0.0 with a subnet mask of 255.255.240.0. Find the following:

- Number of subnets (2 Marks)
- Number of hosts per subnetwork (2 Marks)
- The full range of the last three networks, and (3 Marks)
- The usable address ranges from those last three networks. (3 Marks)

Q5. Consider the network graph given below. Find the shortest routing path from the router A to all other routers using the following routing algorithms and analyze the unused paths and cost in each algorithm.

- Link state routing (5 Marks)
- Distance vector routing (5 Marks)



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	3, 4	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10

CAT - 2 QUESTION PAPER

School	Computer Science and Engineering		
Course Code	CSE3003	Course Title	Computer Networks
Semester	Win 2021	Date of exam	18-04-2022
Slot	A1	Total Marks	50
Duration	90 Min	Class ID	AP2021225000238

Answer all the questions

5 x 10 = 50 Marks

Q1. A sender uses the protocol which can send 3 frames without acknowledgement and can retransmit only the failure frame for reliable transmission of frames. Sender has 10 frames of size 1000 bytes each and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. If every 4th packet that is being transmitted is lost, then how many transactions are required? Explain your answers with timing diagram. (10 Marks)

Q2. Consider a data word 101010.

- Find the codeword using Hamming code algorithm with even parity (5 Marks)
- Check for the bit error if you received the code word as 1011010110 using hamming code when even parity exists (5 Marks)

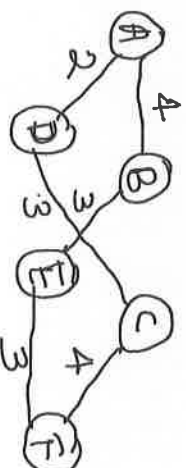
Q3.

- Find the stuffed codeword for the message "This is a message" when the end delimiter is 's' (3 Marks)
- Unstuff the given codeword 101110111011101 when the ED is 1111. (2 Marks)
- Find ICMP checksum when Type = 6, Code = 4, Identifier = 3, Sequence number = 13 and Data = "ABCD" (3 Marks)

- What are addressing mechanisms used in wireless standard of the network layer. (2 Marks)

Q4. The given IP address space is 172.16.192.0 /18. The required number of subnets is 6. Find the subnet table with subnet ID, usable IP range and broadcast range for all 6 subnets. (10 Marks)

Q5. Apply Link state routing and find the shortest path from all the routers to all other routers and draw the unicast routing path to all other nodes for the network graph given below. (10 Marks)



QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q2	3	2,3	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q3	3,4	3,4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q4	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10
Q5	4	4,5	1,2,3,4,5,6,7,8	PEO3,4	PSO2,3	10